Comparative Analysis Report: Baseline vs Naive RAG vs Hybrid RAG

Scope: 50 customer-support queries, three system versions, ROUGE-1/2/L and BLEU

Sources: `Comparative_Results_Full.csv` (per-query scores and answers), `Comparative_Results_Summary.csv` (aggregates)

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1. Executive summary

- On the reported metrics, **both RAG versions score lower than the baseline**, and Hybrid RAG is not meaningfully different from Naive RAG. ROUGE-1 falls from 0.0482 (Baseline) to 0.0312 (Naive RAG) to 0.0274 (Hybrid RAG).
- The baseline-to-Naive drop is statistically significant (paired Wilcoxon p = 0.001 for ROUGE-1). The Naive-to-Hybrid change is **not** (p = 0.23).
- **These numbers should not be read as a quality ranking.** The `reference` column holds intent labels (e.g. `check_payment_methods`), not reference answers, and ROUGE was computed against those labels. Recomputing ROUGE-1 this way reproduces the stored scores exactly. Overlap with a 2-3 word label is a poor proxy for answer quality.
- Reading the answers tells a different story from the scores. RAG answers are far less likely to refuse (30 of 50 baseline answers deflect, versus 9 for Naive and 12 for Hybrid) and often give grounded, specific information. They also show new problems: longer outputs, truncation, and stray hashtag or markdown artifacts.
- **Recommendation:** treat this run as inconclusive on quality. Re-score against gold responses and add a groundedness or LLM-judge metric before deciding between versions (Section 7).

2. Results

2.1 Aggregate scores (mean over 50 queries)

Metric	Baseline	Naive RAG	Hybrid RAG	Baseline → Naive	Naive → Hybrid	Baseline → Hybrid
ROUGE-1	0.0482	0.0312	0.0274	-35.2%	-12.2%	-43.1%
ROUGE-2	0.0083	0.0039	0.0044	-53.4%	+12.7%	-47.5%
ROUGE-L	0.0416	0.0286	0.0262	-31.2%	-8.4%	-37.0%
BLEU	0.00004	0.0	0.0	-100%	undefined	-100%

The summary file matches the means computed from the per-query file.

2.2 Paired significance (per-query differences)

Comparison	Metric	Mean Δ	95% bootstrap CI	Wilcoxon p	Queries better / worse / tied
Baseline → Naive	ROUGE-1	-0.0170	[-0.027, -0.007]	0.001	13 / 30 / 7
Baseline → Naive	ROUGE-L	-0.0130	[-0.022, -0.004]	0.008	16 / 27 / 7
Naive → Hybrid	ROUGE-1	-0.0038	[-0.010, +0.002]	0.225	18 / 21 / 11
Naive → Hybrid	ROUGE-L	-0.0024	[-0.007, +0.002]	0.264	19 / 20 / 11
Baseline → Hybrid	ROUGE-1	-0.0208	[-0.031, -0.011]	<0.001	9 / 34 / 7

Comparison	Metric	Mean Δ	95% bootstrap CI	Wilcoxon p	Queries better / worse / tied
Baseline → Hybrid	ROUGE-L	-0.0154	[-0.024, -0.007]	0.002	12 / 31 / 7

On ROUGE-1, the baseline is the top scorer on 34 of 50 queries (Naive 9, Hybrid 7).

2.3 Notes on specific metrics

- **BLEU is uninformative.** Only one baseline query scores above zero, and no RAG query does. It cannot separate the versions.
- **ROUGE-2 "improvement" for Hybrid (+12.7%)** is noise. The absolute change is 0.0039 to 0.0044, driven by 7 queries up and 5 down.
- All scores are very low in absolute terms (ROUGE-1 below 0.05), which is itself a warning sign about the evaluation setup (Section 3).

3. Why the metrics are misleading

3.1 ROUGE was computed against intent labels

The `reference` column has only 24 distinct values, all intent names such as `place_order`, `track_refund` and `contact_human_agent`. Recomputing ROUGE-1 between each baseline answer and its label matches the stored scores to floating-point precision. So the score measures how many of 1-3 label words appear in a 50-100 word answer.

This has two consequences:

1. **Precision dominates and punishes length.** Mean precision against the label is about 0.026 (Baseline), 0.016 (Naive) and 0.014 (Hybrid). Label-token *recall* is nearly flat at 0.56, 0.53 and 0.53. The versions mention the topic about equally often, and the F1 drop comes mostly from longer answers (mean 50 words for Baseline, 78 for Naive, 85 for Hybrid).
2. **A good answer can score lower than a bad one.** For the query "want assistance to see the payment modalities" (reference `check_payment_methods`):
   - Baseline: apologises that it can't see payment options and tells the user to contact support (ROUGE-1 0.105).
   - Naive RAG: lists cards, digital wallets, bank transfer, gift-card rules and that cash on delivery is unavailable (0.057).
   - Hybrid RAG: the same grounded content, more concisely (0.039).

   The most useful answer gets the lowest score.

3.2 Different evidence from the earlier run

The earlier 130-query file (`rouge_bleu_results.csv`) showed Naive RAG slightly ahead of the baseline (ROUGE-1 0.127 to 0.137). This 50-query run shows the opposite. The two runs use different samples and probably different reference definitions, so neither is reliable on its own. They agree only that ROUGE here is unstable and sensitive to setup.

4. Behavioural findings from the answers

Observation (of 50 answers)	Baseline	Naive RAG	Hybrid RAG
Refusal or deflection ("I'm sorry...", "unable to assist")	30	9	12

Observation (of 50 answers)	Baseline	Naive RAG	Hybrid RAG
Answers containing `{{placeholder}}` / `{%template%}` tokens	15	12	9
Answers containing `#hashtag`-style artifacts	16	30	33
Answers with leaked `User` / `Assistant` dialogue turns	2	3	5
Answers not ending in terminal punctuation (likely truncated)	20	45	47
Queries with ROUGE-1 = 0	12	14	16
Mean answer length (words)	50	78	85

Interpretation:

- **Retrieval clearly changes behaviour.** The baseline mostly refuses or generates template noise. Both RAG versions answer far more often, which is the intended effect of grounding.
- **Hybrid RAG is slightly more verbose and drifts more.** It shows more artifacts and dialogue leakage than Naive RAG. Its only advantage is fewer raw template placeholders. These are small counts on 50 queries and should be treated as directional.
- **Truncation is widespread in the RAG versions** (about 90%). This suggests a generation length cap or missing stop condition and is likely an easy win.

5. Attribution: retrieval versus the Hybrid/fine-tuned component

Contribution	Measured effect (ROUGE-1)	Significant?	Trustworthy as a quality signal?
Retrieval (Baseline → Naive)	-35.2%	Yes (p = 0.001)	No, because of the label-reference issue
Hybrid / fine-tuning (Naive → Hybrid)	-12.2%	No (p = 0.23)	No

Taken literally, neither step helps. The more defensible reading is that **this evaluation cannot attribute improvement to either component.** The qualitative evidence suggests retrieval does help (fewer refusals, specific content), while the evidence for the Hybrid step is weak and mixed: slightly more fluent grounded answers in some cases, but more verbosity and artifacts overall.

6. Limitations

- Only 50 queries, so small effects (like Naive vs Hybrid) cannot be resolved.
- Reference is an intent label, not a gold response.
- BLEU is zero for nearly all rows, and lexical overlap metrics ignore factual correctness and groundedness.
- Answer inspection here was a sample read plus keyword counts, not a systematic human review.
- The artifact counts use simple pattern matching and may over- or under-count slightly.

7. Recommendations

1. **Re-score against gold responses.** Use the dataset's reference answer for each query rather than the intent label, and rerun ROUGE/BLEU.

2. **Add metrics that fit the task:** BERTScore or embedding similarity, an LLM-as-judge rubric for correctness and helpfulness, and a groundedness check that answers are supported by retrieved passages.
3. **Evaluate the router directly.** If Hybrid RAG's value is routing or intent handling, measure intent or route accuracy and retrieval hit rate rather than inferring it from text overlap.
4. **Fix generation settings:** set a sensible max length with proper stop sequences to stop truncation, strip hashtag and `User`/`Assistant` artifacts, and add a post-processing filter for template tokens.
5. **Increase sample size** (at least 150-200 queries, stratified by intent) and report paired bootstrap confidence intervals.
6. **Reconcile the two runs** (130-query and 50-query) by confirming they share the same prompts, references and scoring code.

8. Conclusion

On the current metrics, ranking is Baseline > Naive RAG ≳ Hybrid RAG, with only the Baseline-vs-RAG gap statistically significant. That ranking is an artifact of scoring against intent labels and of RAG answers being longer. The behavioural evidence points the other way for retrieval, and the Hybrid step remains unproven. A corrected evaluation is needed before declaring a winner or crediting fine-tuning with a specific gain.